Eugenio Sotelo

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Education

Florida Polytechnic University

Bachelors of Science in Computer Engineering – GPA: 3.53 Spring 2026
Relevant Skills: Embedded Operating systems, Autonomous Robotic Systems, VLSI Design, Kinematics and Controls of Robotic Systems, Firmware Design, Electric and Hybrid Vehicles, Digital Electronics, Electronic Motor control

Masters of Science in Electrical Engineering – GPA: 3.50 Expected Spring 2027

Experience

Polk County BOCC **June 2024 - August 2024**
Student Intern
- Worked with desktop and hardware support teams to provide troubleshooting and support for customers and citizens of the entire County both in office and on site.
- Worked with mobile support team to deploy, replace, and fix county cell phones and iPads throughout the County.

Phoenix Racing Solar Racing Team **October 2024 – June 2025**
Battery Lead
- Designed and assembled a full high-voltage battery system with 5Kw of power for a solar-powered vehicle competing in competitions across the country, Placing 3rd in the 2025 formula Sun Grand Prix, 5th in the 2026 formula Sun Grand Prix and 6th in the American solar challenge.
- Led custom PCB development, use of the Prohelion BMS with cell-level fusing and monitoring, to deliver a reliable and safe, competition-ready battery system.

Project Work

32-bit CPU Project
Computer Engineering Design Lab
- Designed and built a full 32-bit CPU that supports 10+ instructions as well as the ability to input data and display outputs using system Verilog and the Risc-V architecture.
- Utilized Intel Quartus Prime and the DE10-Lite to upload code and test custom ROMs to confirm features.

Brushless Nerf Blaster Design
Personal project
- Designed and fabricated custom PCBS integrating an RP2040, MOSFET-driven solenoid, and brushless motor control to produce a high-power, competition-ready blaster system Firing 25 darts per second at 220 feet per second.
- Utilized CAD modeling and 3D printing to design and manufacture components to secure electronics integration.

Solenoid Control PCB
Personal Project
- Designed a Circuit that handles the Powering and fast field collapse of the solenoid. An array of P-type and N-type MOSFETs were used in conjunction with Zener Diodes to allow the magnetic field of the solenoid to collapse.
- Utilized a modified half H-bridge design to allow the powering and collapse of the solenoids magnetic field to happen as quickly as possible.
- Allows a solenoid to complete 25 full actuations per second with less than +2° C produced on the PCB and the magnetic field dissipating with ‘regenerative braking’ for the battery.

Extracurricular Activities

Club President
Nerf-Tech
- Manage the entirety of the Club’s E-board and oversaw all activities and events with up to 180 people.
- Continue to run a per-semester HvZ, as well as plan and manage all future events.
- Manage the entirety of the Club’s E-board and oversaw all activities and events.

Skills

Software: Onshape, SolidWorks, Fusion360, Matlab, Microsoft Office, Multisim, Simulink, Arduino IDE, Microchip Studio, Overleaf, C, C++, Firmware Design
Hardware: FDM 3D Printing, DC Circuit Design, PCB Design, Soldering, Oscilloscopes, Function Generators